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28 January 1964
Superseding
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9 April 1956

MILITARY SPECIFICATION

TRIPOD, PHOTOGRAPHIC, PNEUMATIC, METAL

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 SCOPE. - This specification covers the requirements for one type of pneumatic, heavy duty, metal, photographic camera tripod and designated as LM- ( ). (See 6.4).

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and publications of the issue in effect on the date of invitation for bids shall form a part of this specification, to the extend specified herein.

SPECIFICATIONS

Federal

QQ-A-596	Aluminum Alloy, Permanent and Semi-Permanent Mold Castings
QQ-A-601	Aluminum Alloy, Sand Castings
QQ-S-763	Steel Bars, Shapes, and Forgings - Corrosion Resisting

Military

MIL-P-116	Preservation, Methods of
MIL-P-3757	Packaging of Photographic Accessories and Spare Parts, Domestic & Overseas Shipment
MIL-D-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specifications for
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys

FSC 6760

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MIL-M-9868	Microfilming of Engineering Documents, 35mm, Requirements for
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-102	Preservation, Packaging and Packing Levels

Military

MIL-STD-105	Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-152	General Environmental Requirements, Testing Procedures, and Sampling for Aerial Photographic Reconnaissance Equipment
MIL-STD-155	Joint Photographic Type Designation System

(When requesting specifications and other publications, refer to both title and number. Copies of applicable specifications and standards may be obtained upon application to the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120, Attn: Code DCI)

2.2 OTHER PUBLICATIONS. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H-28	Screw Thread Standards for Federal Services

(Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington, D. C. 20402)

AMERICAN STANDARDS ASSOCIATION

PH3.6	Tripod Connections for American Cameras, 1/4 inch - 20 thread

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3.6.1 Vibration. - When the tripod legs are spread and extended, the piston is extended and locked, and a rigidly attached twenty pound weight is mounted on the tripod head, the vibrations caused by the sudden release of a vertical or horizontal stress applied to the tripod head shall be damped in not longer than six seconds when tested as specified in Section 4.

3.6.2 Sturdiness. - The tripod shall withstand continued rough treatment with no loss of function or ease of operation.

3.6.3 Stability. - The stability of the tripod, when tested as specified in Section 4, shall be such that when the legs of the tripod are completely extended and spread and the piston extended and locked, the tripod shall not tip over and shall return to the upright position when any one leg is raised eight inches from the floor and released.

3.7 MATERIAL. - Material shall be as specified herein. All major components shall be made of metal. All non-metallic components shall be made of tough, impact resistant material. All material shall be of good commercial quality entirely suitable for the purpose for which it is used. Material shall be free from all defects and imperfections that might affect the serviceability or appearance of the finished product.

3.7.1 Aluminum Alloy Castings. - Aluminum alloy castings used shall be in accordance with Federal Specifications QQ-A-596 or QQ-A-601.

3.7.2 Fungus Resistance. - Fungus inert materials, as defined in Standard MIL-STD-152, shall be used in the tripod to the maximum extent possible. All other materials shall be suitably finished or treated to resist the growth of fungi.

3.7.3 Metals. - Metals not specified shall be of the corrosion resistant type or shall be suitably protected to resist corrosion from damp salt air during normal service life. Such protection shall in no way prevent compliance with the other requirements of this specification. The use of any protective coating that will crack, chip, or powder with age or extremes of environmental condition shall be avoided.

3.8 CONSTRUCTION

3.8.1 Screw Threads. - Screw threads employed in the tripod shall be of standard commercial pitch and diameter. Screw threads shall conform to the National Bureau of Standards Handbook H-28.

3.8.2 Mating Surfaces. - Metals forming mating surfaces for movable parts shall be chosen so that they will not gall or bind.

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(American Standards may be obtained from the American Standards Association, 10 East 40th Street, New York 16, New York)

3. REQUIREMENTS

3.1 DESCRIPTION. - The tripod shall be an accessory for use with any camera up to and including 8x10 inch view cameras and shall be capable of carrying a weight load of twenty pounds. The tripod shall include a center cylinder, an air piston or post, leg assemblies, and a pan head. The pneumatic device, by which the center post is supported on an air cushion, shall allow the camera to be raised or lowered without friction or jar. The pan (tripod) head shall permit panoramic as well as tilt adjustments.

3.2 DATA. - No data is required by this specification, or by applicable documents referenced in Section 2, unless specified in the contract or order. (See 6.2).

3.3 DIMENSIONS AND WEIGHT

3.3.1 Dimensions. - When all parts are retracted, the tripod shall fold as completely as possible, and shall have envelope dimensions not more than six by six by twenty-five inches exclusive of the pan head. The pan head shall have envelope dimensions of approximately three by seven by nine inches. The measured distance between the legs of the tripod when extended and spread shall be not less than three feet and the height of the tripod, with the legs completely spread and extended and the piston fully extended shall not be less than fifty inches.

3.3.2 Weight. - The weight of the tripod shall not exceed nine pounds.

3.4 INTERCHANGEABILITY. - All similar units and parts of each tripod shall be interchangeable, without modification, with corresponding units and parts of other tripods furnished on the same contract.

3.5 PREPRODUCTION SAMPLE. - Unless otherwise specified (see 6.2) one tripod shall be manufactured using the same methods and procedures as will be used for quantity production and shall be submitted as the preproduction sample for acceptance. Approval of the preproduction sample by the requiring activity shall not relieve the contractor of his obligation to supply equipment conforming to this specification.

3.6 PERFORMANCE. - All parts of the tripod shall work easily and smoothly, and all locking members shall fasten securely and release cleanly.

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3.9.2.2 Collars. - The upper collar shall consist of a fitting permanently locked to the top of the cylinder of the center column. The lower collar shall slide on the cylinder and shall be capable of being locked at any point along its travel.

3.9.2.3 Supporting Members. - Suitable supporting members shall connect the lower collar to the lower end of each of the tripod legs. The supporting members shall be firmly attached to the lower collar and to the tripod legs, and shall allow the tripod legs to be spread uniformly as the lower collar is lowered on the cylinder.

3.9.3 Pan Head. - The pan head shall consist of a camera platform, hinges, a locking collar, and a tripod screw.

3.9.3.1 Camera Platform. - The camera platform shall be a rectangular plate not less than seven by nine inches in area and not less than three eights of an inch thick. The platform shall be an aluminum casting and may be a flat grill formed by a grid of strips having reinforcing ribs. No such strip shall be narrower than a nominal one half inch. The platform shall have three threaded holes along the long axis, one at the center and one near each end, for inserting the tripod screw. Sturdy extensions of the platform shall hinge the platform to the locking collar.

3.9.3.2 Locking Collar. - The locking collar shall fit and clamp upon the head of the piston of the center column. The locking collar shall have sturdy extensions which will hinge the collar to the camera platform. The locking collar shall allow the pan head to be rotated freely in any direction or to be removed from the head of the piston.

3.9.3.3 Hinge Locking Mechanism. - The extensions of the camera platform and the locking collar shall constitute a hinge and locking device. The mechanism to lock the hinge shall allow the hinge to move freely when unlocked and shall resist a pressure of twenty pounds applied to the rear of the camera platform when the hinge is locked in any position. The rear of the camera platform shall be considered to be the end opposite to the hinge locking mechanism. The mechanism shall allow sufficient leverage for the operator to lock the mechanism without undue exertion. The hinge shall be sufficiently strong to withstand the stresses involved. The hinge shall allow the camera platform at least 90 degrees of rotation, including a vertical and a horizontal position.

3.9.3.4 Tripod Screw. - The tripod screw shall conform to American Standard PH3.6 and shall have the 0.340 inch maximum length listed therein. The tripod screw shall be readily

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3.8.3 Rigidity and Alignment. - Each component of the tripod shall be of sufficient strength and rigidity to accomplish its function and to maintain smooth and easy operation of the tripod.

3.9 COMPONENTS. - The components of the tripod shall be a center column, a leg assembly, and a pan head.

3.9.1 Center Column. - The center column shall consist of a metal cylinder, a piston, and a valve.

3.9.1.1 Piston. - The length of the piston shall be sufficient to allow the tripod to be extended to not less than 50 inches. In no case shall the length of the piston be less than 20 inches. A cushion of air shall prevent the piston from dropping suddenly to the bottom of the cylinder. The piston shall at all times extend out of the cylinder far enough to allow the pan head to be attached properly. The bottom of the cylinder shall carry an airtight fitting which shall resist abrasion and shall not damage the cylinder. The top of the piston shall be closed.

3.9.1.2 Cylinder. - The cylinder shall be of an appropriate size to receive the piston. The walls of the cylinder shall be sufficiently thick to resist deformation by the lower collar of the leg assembly as the collar is locked in various positions.

3.9.1.3 Valve. - The valve shall be mounted at the bottom of the cylinder and shall allow air to enter the cylinder freely. The valve shall restrict the outflow of air as the piston descends so that the piston shall not reach the bottom of its travel in less than three seconds from an elevation of one foot when a twenty pound weight is attached to the pan head, or in more than forty seconds from full elevation with no weight attached to the pan head.

3.9.2 Leg Assembly. - The leg assembly shall consist of three legs with one extension tube each, two collars, and suitable supporting members.

3.9.2.1 Legs. - The three tripod legs shall be firmly hinged to the upper collar. Each leg shall contain a telescoping extension tube which can be locked at any point in its travel by means of a knurled knob. Suitable stops shall insure that the extensions do not retract far enough to allow the center column to project beyond them.

3.9.2.1.1 Tips. - Each extension tube shall be provided with a reversible steel tip in accordance with Class 302 of Federal Specification QQ-S-763. The tip shall be furnished with a point at one end and a rounded piece of rubber or suitable plastic at the other end. The tip shall fit snugly into the extension tube and shall be easily removable regardless of which end is inserted into the extension tube.

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accessible at any position of the pan head and shall not interfere with the travel of the camera platform. The tripod screw shall be of the removable captive type.

3.9.4 Control Knobs. - The control knobs exclusive of those which lock the leg extensions shall be a nominal one inch in length and a nominal one and a half inches in diameter. The knobs which lock the leg extensions shall be knurled for easy grasping and shall not be less than one half inch in diameter. The control knobs and their shafts shall be so arranged that they do not interfere with each other or with the rotation of the pan head. The knobs shall be securely attached to their shafts. The control knob shafts shall be captive.

3.10 ENVIRONMENTAL CONDITIONS

3.10.1 Operating Environmental Conditions. - The tripod shall operate satisfactorily in any ambient temperature from minus 40 to plus 120 F. at any relative humidity up to 90 percent.

3.10.2 Storage Environmental Conditions. - Storage of the tripod at any ambient temperature from minus 65 to plus 160 F at any relative humidity up to 90 percent shall have no deleterious effect on any component. After exposure all parts shall operate smoothly and easily.

3.11 COLOR AND FINISH. - All external aluminum surfaces shall be given a clear anodized finish in accordance with Military Specification MIL-A-8625. If the tripod or any part thereof is painted, the color shall be suitable and the paint shall be durable and scuff-resistant. The tripod shall be free of sharp angles, burrs, projections, etc., which might constitute a hazard to the operator.

3.12 IDENTIFICATION MARKING. - Identification marking shall be in accordance with MIL-STD-130.

3.13 DRAWINGS. - When specified in the procurement document (see 6.2) a complete set of drawings and all associated lists conforming to Specification MIL-D-70327 shall be furnished.

3.13.1 Microfilm of Drawings and Related Data. - When specified in the procurement document (see 6.2), microfilm conforming to Specification MIL-M-9868 shall be made of the drawings and associated lists.

3.14 WORKMANSHIP. - Workmanship shall conform to good commercial practice in the fabrication of tripods.

4. QUALITY ASSURANCE PROVISIONS

4.1 RESPONSIBILITY FOR INSPECTION. - Unless otherwise specified in the contract or order (see 6.2), the supplier is

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responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified (see 6.2), the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 TEST CONDITIONS AND MATERIAL. - Unless otherwise specified herein, tests shall be conducted under ordinary room conditions of temperature and humidity. The equipment and supplies necessary for testing shall be furnished by the contractor

4.3 CLASSIFICATION OF INSPECTIONS. - Inspections shall be listed in Table I and shall be classified as follows:

(a) Preproduction Inspection
(b) Quality Conformance Inspection
(c) Preparation for Delivery Inspection

TABLE I

EXAMINATIONS AND TESTS

Tests	Requirement	Test Method
Preproduction Sample Inspection		
Visual and Mechanical Examination	3.3, 3.3.1, 3.3.2, 3.7, 3.11, 3.12, 3.14	4.9.1
Interchangeability Test	3.4	4.9.2
Performance Test	3.6	4.9.3
Rigidity Test	3.8.3	4.9.4
Sturdiness Test	3.6.2	4.9.5
Stability Test	3.6.3	4.9.6
Control Knob Test	3.9.4	4.9.7
Valve Test	3.9.1.3	4.9.8
Cylinder Deformation Test	3.9.1.2	4.9.9
Hinge Locking Mechanism Test	3.9.3.3	4.9.10
Vibration Test	3.6.1	4.9.11
Fungus Resistance Test	3.7.2	4.9.12
Operating Environment Test	3.10.1	4.9.13
Storage Environment Test	3.10.2	4.9.14
Quality Conformance Inspection		
Visual and Mechanical Examination	3.3, 3.3.1 3.3.2, 3.7, 3.11, 3.12, 3.14	4.9.1

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4.6.1 Sampling for Preparation for Delivery Inspection. - Sampling procedures shall be in accordance with Standard MIL-STD-105. The inspection level shall be I and the acceptable quality level (AQL) shall be 1.0 percent for major defects and 4.0 percent for minor defects.

4.7 LOT. For the purpose of sampling, a lot shall be defined as follows:

4.7.1 Quality Conformance Inspection Lot. - For quality conformance inspection, a lot shall consist of those tripods assembled, cleaned, and preserved, which are ready for packaging as a single shipment.

4.7.2 Preparation for Delivery Inspection Lot. - For preparation for delivery inspection, a lot shall consist of those tripods packaged and packed for shipment.

4.8 CLASSIFICATION OF DEFECTS. - Classification of defects shall be as defined in Military Standard MIL-STD-105.

4.8.1 Quality Conformance Inspection Defects. -

Critical Defects

1. Pan head not interchangeable with other tripods.

Major Defects

101. Tripod not the same as preproduction sample.
102. Parts missing - fasteners, screws, rivets, knobs, leg-tips, etc.
103. Extension legtips not interchangeable with same or other tripods.
104. Operation of tripod not smooth.
105. Valve inoperative or damaged.
106. Control knobs loose, damaged, not captive, shafts bent or stripped.
107. Difficulty in locking or unlocking collars.
108. Dimensions not as specified.
109. Weight not as specified.

Minor Defects

201. Scratches, cuts, abrasions of finish.
202. Identification label missing, loose or damaged.
203. Inferior workmanship - sharp edges, burrs, etc.
204. Tripod and pan head improperly cleaned.
205. Preservative missing or improperly applied.

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TABLE I (Cont'd.)

EXAMINATIONS AND TESTS

Tests	Requirement	Test Method
Interchangeability Examination	3.4	4.9.2
Performance Test	3.6	4.9.3
Valve Test	3.9.1.3	4.9.8
Packaging and Packing Inspection		
Packaging	3.14, 5.1.1.4	4.9.15
Packing	5.2	4.9.15
Marking	5.3	4.9.15

4.4 PREPRODUCTION INSPECTION. - Prior to submission for acceptance, the preproduction sample shall be subjected by the contractor to the preproduction inspection specified herein. On completion of the inspection by the contractor, the preproduction sample shall be shipped to the Military Testing Agency designated by the requiring activity (see 6.2). Unless otherwise specified (see 6.2) a report of the results of the tests shall accompany the preproduction sample. The Military Testing Agency may repeat the preproduction inspection or make other tests as needed to verify that the production methods and choice of design details are capable of yielding equipment that complies with the requirements of this specification. Approval by the requiring activity shall be based on no defects in the preproduction sample.

4.5 QUALITY CONFORMANCE INSPECTION. - Each production tripod, subsequent to the preproduction sample, submitted for acceptance under the contract or order, shall be subjected by the contractor to the quality conformance inspection specified herein. The quality conformance inspection shall verify that all subsequent tripods are duplicates of the approved preproduction sample.

4.5.1 Sampling for Quality Conformance Inspection. - Sampling procedures shall be in accordance with Military Standard MIL-STD-105. The inspection level shall be I and the acceptable quality level (AQL) shall be 1.0 percent for major defects and 4.0 percent for minor defects.

4.6 PREPARATION FOR DELIVERY INSPECTION. - Examine the preservation, packaging, packing and marking for shipment and storage in accordance with the preparation for delivery inspection specified herein.

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4.8.2 Preparation for Delivery Inspection Defects. -

Critical Defects

None Defined

Major Defects

110. Use of improper or defective material.
111. Quantity in pack not as specified.
112. Incorrect packaging method applied.
113. Cushioning or padding omitted.
114. Cushioning inadequate for physical and mechanical protection of the item.
115. Gross weight in excess of specified amount.
116. Box closure not as specified.
117. Type, grade, class, and style of shipping container not as specified.
118. Strapping, if applicable, omitted, inadequate or improperly applied.
119. Marking omitted, incorrect, illegible, or not in accordance with MIL-STD-129.
120. Shipping documents or packing list omitted.

Minor Defects

206. Unsealed unit package.
207. Defective taping or sealing of unit package.
208. Any other defects which may be considered minor by definition of Standard MIL-STD-105.

4.9 TESTS AND EXAMINATIONS

4.9.1 Visual and Mechanical Examination. - Examine the tripod visually and mechanically to determine that it is a duplicate of the tripod which passed the preproduction inspection and with respect to completeness, cleanliness, performance and workmanship.

4.9.2 Interchangeability Examination. - The dimensions listed below shall be gauged or measured to determine compliance with the physical interchangeability requirements. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of component parts and insertable items when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

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The acceptance test of the assembled item, shall consist of conplete interchangeability of the pan head and the extension leg tips between units. Non-interchangeability shall be cause for rejection.

4.9.3 Performance Test. - Extend and retract each leg, reverse each extension tip, lock and unlock each locking collar, and lock and unlock the hinge locking mechanism of the pan head, ten times each. All parts shall operate freely and easily, all locking adjustments shall lock positively without requiring excessive force, and the pan head shall be capable of locking at any random combination of tilt and turn.

4.9.4 Rigidity Test. - Extend and spread the tripod legs. Raise and lock the piston to its maximum height. Drop a twenty pound weight on the camera platform from a distance of not less than 1-1/2 inches nor more than two inches. Repeat ten times. There shall be no bowing of legs, looseness of joints and fittings, and the locking mechanisms shall support the weight without slipping.

4.9.5 Sturdiness Test. - With the tripod in its retracted and locked condition, extend it parallel to a concrete floor and drop it from a height of a nominal two feet. Repeat the test ten times. The tripod shall operate smoothly and easily after the test. There shall be no fractures on the castings and no evidence of loss of rigidity or function in any part of the tripod.

4.9.6 Stability Test. - Extend and spread the tripod legs and raise and lock the piston at its maximum height. Raise one leg a nominal 8 inches from the floor and release it. Repeat the test with a compact twenty pound weight attached to the center of the camera platform. With or without the weight attached, the tripod shall return to the upright position.

4.9.7 Control Knob Test. - Grasp each control knob in turn and verify that it is of an appropriate size and shape to allow the operator to hold and turn the knob firmly. Using the hand only, attempt to turn each control knob beyond its firmly locking point, then turn in the other direction until the knob moves freely. Test the control knob for captivity. Repeat ten times. No knob shall become loose on its shaft, no fitting shall have been damaged and the knob is captive and not easily removable.

4.9.8 Valve Test. - Set up the tripod in the fully extended position and lock the lower column. Raise the piston to the top of its travel without locking and time the descent of the piston to the bottom of its travel. The time of travel shall not exceed forty seconds. Attach a 20 pound weight to the center of the camera platform and raise the piston one foot from its minimum extension. The piston when released shall require not less than 3 seconds to reach the bottom of its travel.

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4.9.13 Operating Environment Test. - Place the tripod in a test chamber and lower the temperature to minus 40 F. At the end of the four-hour period conduct the performance test. Raise the temperature in the test chamber to 120 F and increase the humidity gradually over a four-hour period from 10 percent or less to 90 percent. At the end of the four-hour period conduct the performance test.

4.9.14 Storage Environment Test. - Place the tripod in a test chamber and lower the temperature to minus 65 F. Over a 24-hour period, cycle the humidity at least once to include a combination which shall cause moisture to condense on the tripod. At the end of the 24-hour period, raise the temperature to 160 F. and cycle the humidity from 10 percent or less to 90 percent over a 24-hour period. After the tripod has returned to room temperature, it shall be carefully examined for signs of damage and for ability to pass the performance test.

4.9.15 Packaging and Packing Examination. - Examine the preservation, packaging, packing and marking for shipment and storage visually and mechanically to determine conformance to the requirements of the applicable specifications.

5. PREPARATION FOR DELIVERY

5.1 PRESERVATION AND PACKAGING. - Preservation and packaging shall be Level A or C as defined in Federal Standard No. 102 and as specified by the requiring activity. The tripod shall be packaged as one integral unit (see 6.2).

5.1.1 Level A. - The tripod shall be preserved and packaged in accordance with the procedures specified (for the designated methods as prescribed) in Specification MIL-P-116 and described in the following paragraphs.

5.1.1.1 Cleaning. - All unpainted metal surfaces on each tripod shall be cleaned following the procedures specified in Specification MIL-P-116 for C-1 cleaning.

5.1.1.2 Drying. - Cleaned metal surfaces shall be dried in accordance with procedures specified in D-4, wiping as prescribed in Specification MIL-P-116.

5.1.1.3 Preservative. - Preservative shall be type P-2 or P-9 as specified in Specification MIL-P-116. Preservatives shall not be applied to any painted surfaces.

5.1.1.4 Packaging. - The tripod shall be packaged Method 1A-2 as prescribed in Specification MIL-P-116.

5.1.2 Level C. - The tripod shall be packaged in accordance with commercial practice and in a manner that will afford

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4.9.9 Cylinder Deformation Test. - Set up the tripod in the fully extended position without locking the lower collar. Raise the piston with no weight on the camera platform and allow it to descend. Record the time required, to the nearest second, for the piston to complete its descent to the fully lowered position. Repeat ten times. The longest descent time shall be designated the maximum unlocked descent time. Lock the lower collar and allow the piston to descend from the fully raised position. Move the lower collar to at least six other positions up the center column and repeat the test at each position. The descent time of the piston in each test shall not exceed the maximum unlocked descent time.

4.9.10 Hinge Locking Mechanism Test. - Mount a 20 pound weight on the rear of the camera platform, tilt the camera platform to approximately fifteen degrees and lock the hinge mechanism. Repeat at 30, 45, 60, and 75 degrees. Measuring from a horizontal rigidly attached to the upper collar of the leg assembly, the deflection of the front edge of the camera platform shall not exceed 1/8 inches after being locked at any setting.

4.9.11 Vibration Test. - Extend and spread the tripod legs and raise and lock the piston at its maximum elevation. Mount on the camera platform a still projector together with sufficient rigidly attached weight to total twenty pounds. The projector should have a focal length of a nominal four inches. Set up the tripod on a concrete floor in as vibration-free an environment as possible. Project on a screen twelve feet from the film plane of the projector, with one tripod leg pointing in the direction of the optical axis. If no four inch lens is available, set up the projector with an available lens and place the screen thirty-six focal lengths from the projector film plane. Project upon the screen the image of cross-hairs in which the lines have a projected thickness of 1/64 of an inch or less. Press down upon the rear portion of the pan head with sufficient force to displace the image of the cross-hairs on the screen by approximately 1/4 inch and release sharply. Repeat four times. The arithmetical average of the elapsed time for image motion to damp to oscillations 1/32 inch or less, displaced from the position of the cross-hairs at rest, shall not exceed six seconds. Press the rear of the pan head to one side with sufficient force to displace the image of the cross-hairs approximately 1/4 inch laterally and release sharply. The vibrations shall damp to the above defined limits within an arithmetical average of six seconds for five measurements.

4.9.12 Fungus Resistance Test. - If any fungus nutrient materials as defined in MIL-STD-152 are used in the tripod, the fungus resistance test, Procedure I, of Specification MIL-E-5272 shall be applied to such materials except that the period shall be fourteen days. The result of the test shall be evaluated according to MIL-E-5272. If the materials fail the test, the entire tripod shall be considered to have failed the test.

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protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 PACKING. - The levels of packing shall be as specified by the procuring activity. (See 6.2)

5.2.1 Level A. - Unit packages shall be packed for shipment in accordance with Specification MIL-P-3757. No more than twenty unit packages shall be packed in any one shipping container.

5.2.2 Level C. - Unit packages shall be packed for shipment in accordance with Specification MIL-P-3757. No more than twenty unit packages shall be packed in any one wood shipping container. If fiberboard shipping containers are used, the number of unit packages packed shall be such as to insure safe delivery to the point of acceptance.

5.3 MARKING. - All unit packages and shipping containers shall be marked in accordance with Military Standard MIL-STD-129.

6. NOTES

6.1 INTENDED USE. - The tripod is intended for use with cameras up to and including 8 x 10 inch view cameras with a maximum weight of 20 pounds for studio and field photography.

6.2 ORDERING DATA. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Quantity desired.
(c) Data required (see 3.2, 3.13, 3.13.1, and 4.4).
(d) Preproduction sample, if not required (see 3.5 and 6.3).
(e) Responsibility for Inspection (see 4.1).
(f) Designation of Military Testing Agency (see 4.4).
(g) Level of preservation, packaging and packing (see Section 5).

6.3 PREPRODUCTION. - If the tripod has passed the preproduction inspection during the past twelve months, the requirement for a preproduction sample of additional, identical tripods may be waived. Authorization to omit the preproduction sample may be requested by the contractor from the requiring activity.

6.4 TYPE DESIGNATOR. - In accordance with the provisions of MIL-STD-155, the parenthesis in the type designator will be replaced by a letter, number, or both. This information will

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be furnished upon application by the contractor to the requiring activity after award of each contract. The complete type designator shall then be used on nameplates, in instruction books, and wherever reference is made to the equipment covered by this specification.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-T-19478A(Wep)-Tripod, Photographic, Pneumatic, Metal

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426

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DEPARTMENT OF THE NAVY

Bureau of Naval Weapons
Washington, D. C. 20360

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